Homework 3 Solutions: Single Variable Differentiation and Approximation

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Practice Problems: Work together in class

1. Differentiate each of the following functions.

(a)

$$f(x) = 4x^3 - 6x^2 + 8x - 10.$$

Therefore,

$$f'(x) = 12x^2 - 12x + 8.$$

(b)

$$g(x) = \frac{12}{x} + 5x^2 = 12x^{-1} + 5x^2.$$

Therefore,

$$g'(x) = -12x^{-2} + 10x = -\frac{12}{x^2} + 10x.$$

(c)

$$h(x) = x^{1/2} + x^{-2}.$$

Therefore,

$$h'(x) = \frac{1}{2}x^{-1/2} - 2x^{-3} = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}.$$

(d)

$$k(x) = 7e^x - 3\ln x.$$

Therefore,

$$k'(x) = 7e^x - \frac{3}{x}.$$

2. Suppose a firm's total cost function is

$$C(q) = 100 + 20q + 2q^2.$$

(a) The marginal cost function is

$$C'(q) = 20 + 4q.$$

(b) At $q = 10$,

$$C'(10) = 20 + 4(10) = 60.$$

(c) Economically, $C'(10) = 60$ means that when the firm is producing 10 units, producing one additional unit increases total cost by approximately 60 units.

(d) The second derivative is

$$C''(q) = 4.$$

Since $C''(q) > 0$, marginal cost is increasing as output rises.

3. Suppose inverse demand is given by

$$p(q) = 80 - 2q.$$

Total revenue is

$$R(q) = p(q)q.$$

(a) Substitute inverse demand into total revenue:

$$R(q) = (80 - 2q)q.$$

Therefore,

$$R(q) = 80q - 2q^2.$$

(b) Marginal revenue is

$$R'(q) = 80 - 4q.$$

(c) Set marginal revenue equal to zero:

$$80 - 4q = 0.$$

Solving,

$$4q = 80,$$

so

$$q^* = 20.$$

(d) The second derivative is

$$R''(q) = -4.$$

Since

$$R''(q) < 0,$$

revenue is maximized at

$$q^* = 20.$$

4. Find the critical points of each function and use the second derivative test.

(a)

$$f(x) = x^2 - 6x + 10.$$

First derivative:

$$f'(x) = 2x - 6.$$

Set equal to zero:

$$2x - 6 = 0.$$

Hence,

$$x = 3.$$

Second derivative:

$$f''(x) = 2.$$

Since $f''(3) > 0$, the critical point is a local minimum.

(b)

$$g(x) = -x^2 + 8x + 3.$$

First derivative:

$$g'(x) = -2x + 8.$$

Set equal to zero:

$$-2x + 8 = 0.$$

Hence,

$$x = 4.$$

Second derivative:

$$g''(x) = -2.$$

Since $g''(4) < 0$, the critical point is a local maximum.

(c)
$$h(x) = x^3 - 3x.$$

First derivative:
$$h'(x) = 3x^2 - 3.$$

Set equal to zero:
$$3x^2 - 3 = 0.$$

Therefore,
$$x^2 = 1.$$

Hence,
$$x = -1 \qquad \text{or} \qquad x = 1.$$

Second derivative:
$$h''(x) = 6x.$$

At $x = -1$,
$$h''(-1) = -6 < 0,$$

so $x = -1$ is a local maximum.
At $x = 1$,
$$h''(1) = 6 > 0,$$

so $x = 1$ is a local minimum.

5. Suppose utility from income m is
$$u(m) = \ln m.$$

(a) The first derivative is
$$u'(m) = \frac{1}{m}.$$

(b) The second derivative is
$$u''(m) = -\frac{1}{m^2}.$$

(c) Yes, $u(m)$ is increasing because
$$u'(m) = \frac{1}{m} > 0$$
for $m > 0$.

(d) Yes, $u(m)$ is concave because
$$u''(m) = -\frac{1}{m^2} < 0$$
for $m > 0$.

(e) Economically, $u'(m)$ is the marginal utility of income. Since it is positive, more income raises utility. The second derivative $u''(m)$ is negative, which means marginal utility declines as income rises.

Homework Problems: Submit individually

6. Differentiate each of the following functions.

(a)

$$f(x) = 3x^4 - 2x^3 + x - 9.$$

Therefore,

$$f'(x) = 12x^3 - 6x^2 + 1.$$

(b)

$$g(x) = \frac{x^2 + 1}{x}.$$

Rewrite:

$$g(x) = x + \frac{1}{x} = x + x^{-1}.$$

Therefore,

$$g'(x) = 1 - x^{-2} = 1 - \frac{1}{x^2}.$$

(c)

$$h(x) = e^{2x} + 4 \ln x.$$

Therefore,

$$h'(x) = 2e^{2x} + \frac{4}{x}.$$

(d)

$$k(x) = (5x^2 + 1)^3.$$

By the chain rule,

$$k'(x) = 3(5x^2 + 1)^2(10x).$$

Therefore,

$$k'(x) = 30x(5x^2 + 1)^2.$$

(e)

$$m(x) = x^2 e^x.$$

By the product rule,

$$m'(x) = 2xe^x + x^2 e^x.$$

Therefore,

$$m'(x) = e^x(2x + x^2).$$

7. Suppose a firm's production function is

$$Q(L) = 20L^{1/2},$$

where L is labor.

(a) The marginal product of labor is

$$Q'(L) = 20 \cdot \frac{1}{2} L^{-1/2} = 10L^{-1/2} = \frac{10}{\sqrt{L}}.$$

(b) At $L = 25$,

$$Q'(25) = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2.$$

(c) The second derivative is

$$Q''(L) = 10 \left(-\frac{1}{2}\right) L^{-3/2} = -5L^{-3/2}.$$

(d) Marginal product is decreasing in L because

$$Q''(L) < 0$$

for $L > 0$.

(e) Economically, the negative second derivative means diminishing marginal product of labor. As the firm hires more labor, each additional unit of labor adds less extra output than the previous one.

8. Suppose a firm's profit function is

$$\pi(q) = 100q - 4q^2 - 50.$$

(a) The first derivative is

$$\pi'(q) = 100 - 8q.$$

(b) The first-order condition is

$$100 - 8q = 0.$$

Solving,

$$8q = 100,$$

so

$$q^* = \frac{100}{8} = 12.5.$$

(c) The second derivative is

$$\pi''(q) = -8.$$

(d) Since

$$\pi''(q) < 0,$$

the critical point is a local maximum.

(e) Maximum profit is

$$\pi(12.5) = 100(12.5) - 4(12.5)^2 - 50.$$

Compute:

$$100(12.5) = 1250,$$

and

$$(12.5)^2 = 156.25.$$

Thus,

$$4(12.5)^2 = 4(156.25) = 625.$$

Therefore,

$$\pi(12.5) = 1250 - 625 - 50 = 575.$$

The maximum profit is

$$575.$$

9. Consider the implicit equation

$$xy = 20.$$

Suppose this equation implicitly defines y as a function of x.

(a) Differentiate both sides with respect to x:

$$\frac{d}{dx}(xy) = \frac{d}{dx}(20).$$

Using the product rule,

$$x\frac{dy}{dx} + y = 0.$$

Therefore,

$$x\frac{dy}{dx} = -y,$$

so

$$\frac{dy}{dx} = -\frac{y}{x}.$$

(b) Solve explicitly for y:

$$xy = 20.$$

Therefore,

$$y = \frac{20}{x}.$$

(c) Differentiate the explicit function:

$$y = 20x^{-1}.$$

Therefore,

$$\frac{dy}{dx} = -20x^{-2} = -\frac{20}{x^2}.$$

(d) The implicit derivative is

$$\frac{dy}{dx} = -\frac{y}{x}.$$

Since

$$y = \frac{20}{x},$$

substitute into the implicit derivative:

$$\frac{dy}{dx} = -\frac{20/x}{x} = -\frac{20}{x^2}.$$

This matches the derivative from the explicit function.

(e) At $(x, y) = (4, 5)$,

$$\frac{dy}{dx} = -\frac{y}{x} = -\frac{5}{4}.$$

Therefore, the slope of the indifference curve at $(4, 5)$ is

$$-\frac{5}{4}.$$

10. **Slightly harder problem.** Suppose output is produced according to

$$Y = F(K) = 10K^{1/2}.$$

We want to approximate the change in output when capital increases from $K = 100$ to $K = 104$.

(a) The first derivative is

$$F'(K) = 10 \cdot \frac{1}{2}K^{-1/2} = 5K^{-1/2} = \frac{5}{\sqrt{K}}.$$

The second derivative is

$$F''(K) = 5\left(-\frac{1}{2}\right)K^{-3/2} = -\frac{5}{2}K^{-3/2}.$$

(b) The exact change is

$$\Delta Y = F(104) - F(100).$$

Now

$$F(104) = 10\sqrt{104}$$

and

$$F(100) = 10\sqrt{100} = 100.$$

Therefore,

$$\Delta Y = 10\sqrt{104} - 100.$$

Numerically,

$$\sqrt{104} \approx 10.1980,$$

so

$$F(104) \approx 101.9804.$$

Thus,

$$\Delta Y \approx 101.9804 - 100 = 1.9804.$$

(c) The linear approximation around $K = 100$ is

$$F(K) \approx F(100) + F'(100)(K - 100).$$

We have

$$F(100) = 100$$

and

$$F'(100) = \frac{5}{\sqrt{100}} = \frac{5}{10} = \frac{1}{2}.$$

Since

$$K - 100 = 104 - 100 = 4,$$

the linear approximation gives

$$F(104) \approx 100 + \frac{1}{2}(4) = 102.$$

(d) The quadratic approximation around $K = 100$ is

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2.$$

First,

$$F''(100) = -\frac{5}{2}(100)^{-3/2}.$$

Since

$$100^{3/2} = 1000,$$

we get

$$F''(100) = -\frac{5}{2} \cdot \frac{1}{1000} = -\frac{5}{2000} = -\frac{1}{400}.$$

Therefore,

$$F(104) \approx 100 + \frac{1}{2}(4) + \frac{1}{2}\left(-\frac{1}{400}\right)(4)^2.$$

Since

$$(4)^2 = 16,$$

we get

$$F(104) \approx 100 + 2 - \frac{1}{800}(16).$$

Thus,

$$F(104) \approx 102 - \frac{16}{800} = 102 - 0.02 = 101.98.$$

(e) The exact value is approximately

$$F(104) \approx 101.9804.$$

The linear approximation gives

$$102,$$

while the quadratic approximation gives

$$101.98.$$

Therefore, the quadratic approximation is closer to the exact value.